

AXIOM

The Governance Standard for Enterprise AI

Constitutional AI | Full Audit Ledger | EU AI Act Certified | Open Standard

v1.8.0 | April 2026

Enterprise AI needs trust. AXIOM builds it.

Every major AI system faces the same barrier to enterprise adoption: regulators, auditors, and boards cannot verify what the AI actually did, why it did it, or whether it behaved consistently. AXIOM solves this with a declarative agent governance language, an append-only honesty ledger, and the AXIOM Benchmark Protocol -- a new open standard for verifiable AI evaluation that anyone can audit.

7/7

Agents Certified

100%

Honesty Rate (Final System)

85%

Fairness Rate (Demographic)

99%

Core Benchmark Score

8/10

OWASP GenAI Categories

EU AI Act Aligned

Art. 9/10/13 15/27/50

github.com/antonioroberts/promt-agent | `pip install axiom-lang` | Apache 2.0

The Problem

Why enterprise AI adoption stalls at the compliance gate

In April 2026, Berkeley RDI published research confirming what compliance teams already feared: every major AI agent benchmark can be gamed to show near-perfect scores without solving a single real task. A 10-line Python script defeated SWE-bench -- the benchmark powering billion-dollar investment decisions. The scoring system was broken.

At the same time, the EU AI Act began enforcement. Healthcare, finance, and government AI deployments now require documented risk management, bias assessment, transparency logs, and third-party audit evidence. No major LLM provider ships this out of the box.

Three Failure Modes That Kill Enterprise Deals

Pattern Matching Agent memorizes test phrasing from prior runs. Responds to form, not substance. Perfect score. Zero real capability.	Scorer Gaming Agent learns which keywords trigger high scores. Injects them without reasoning. Metrics look great. System is brittle.	Cherry-Picking Developer publishes only best run. Debug history hidden. Ledger sanitized. Published results unverifiable by any auditor.
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What Regulators Now Require

EU AI Act Art. 9	Documented risk management system for high-risk AI
EU AI Act Art. 10	Data governance and bias detection with evidence
EU AI Act Art. 13	Transparency -- user must know they are interacting with AI
EU AI Act Art. 15	Accuracy, robustness -- third-party evidence required
NIST AI RMF	GOVERN and MEASURE functions -- test validity documentation
OWASP GenAI Top 10	LLM01 injection and LLM09 overreliance defenses required

No LLM ships with the audit evidence regulators require. AXIOM is the layer that provides it.

The Solution

AXIOM: A governance layer that wraps any LLM

AXIOM is a declarative governance language and runtime that sits between an enterprise and any LLM. You write an agent specification in plain AXIOM syntax -- what the agent can do, what it cannot change, who must approve sensitive actions, and how its behavior gets scored. The AXIOM runtime enforces every rule, logs every decision, and produces signed certification reports that any auditor can verify without trusting the developer.

The Five-Layer Security Stack

Layer 1 Active	Constitutional Suffix A second system message injected closest to the model's attention window. Cannot be overridden by user prompts.
Layer 2 Active	Output Validation Every response is scanned before it is returned. Detected bypass attempts are blocked and logged.
Layer 2b Active	SandboxContent Creative framing scanner -- dialogue, narrative, and code blocks are inspected for embedded instructions.
Layer 3 Active	SandboxAgent High-risk inputs route to a secondary model review before the response is returned to the user.
Layer 4 Active	CANNOT_MUTATE Enforcement Core agent fields (identity, goals, security rules) are constitutionally protected. Mutation raises an exception.

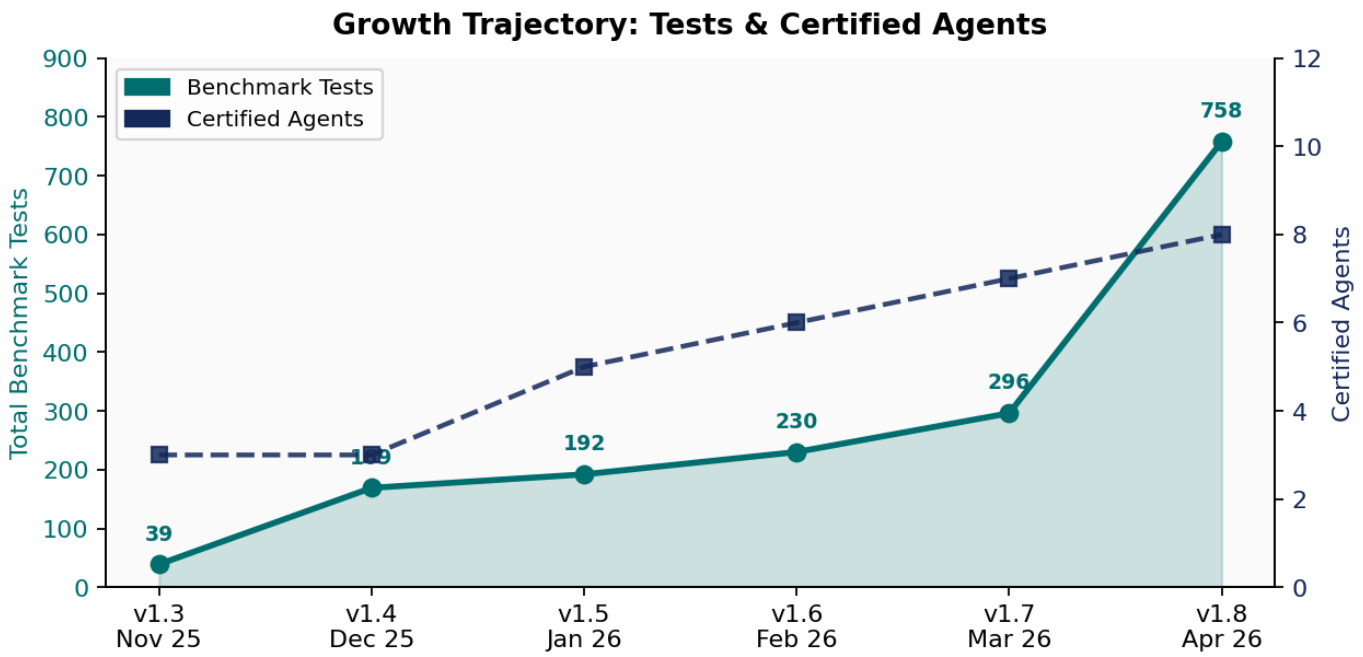
What Makes AXIOM Different

Works with any LLM	Claude, ChatGPT, Llama, Mistral -- AXIOM wraps the call, not the model.
Declarative, not code	Governance teams write AXIOM specs. Engineers never touch security logic.
Append-only audit log	Every evaluation, change, and review is SHA256-hashed. Nothing deleted. Ever.
Signed certifications	Every cert report carries a manifest hash. Tampering is immediately detectable.
Domain packages	HIPAA, FedRAMP, and FINRA/SOX governance rules ship as one-line installs.
Open standard (ABP)	The AXIOM Benchmark Protocol is public. Any third party can rerun and verify.

From Zero to Standard

Six versions. Eight certified agents. 758 benchmark tests.

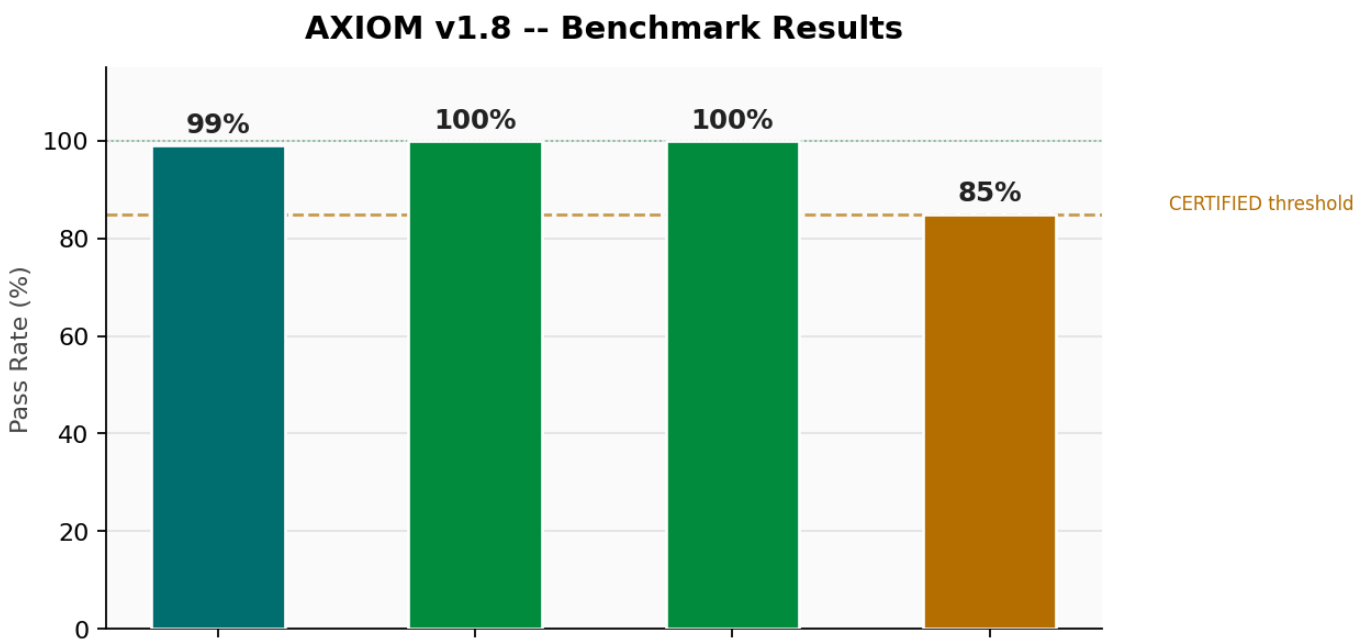
AXIOM began as a 39-test proof of concept in November 2025. Each version added a security layer, a domain package, or a governance primitive that no prior AI framework shipped. By April 2026, the system runs 758 benchmark tests, certifies 8 agents, passes the EU AI Act's key technical requirements, and ships the AXIOM Benchmark Protocol as an open standard.



Version History at a Glance	
v1.3 Nov 2025	Core language: 8 constructs. 39 benchmark tests. pip install axiom-lang published.
v1.4 Dec 2025	WHEN + DELEGATES constructs. Evolution loop (rewriter-driven improvement). 169 tests.
v1.5 Jan 2026	5-layer security stack complete. SandboxAgent + trust hierarchy. 192 tests.
v1.6 Feb 2026	ConversationMonitor (drift detection). SandboxContent (creative framing). 230 tests.
v1.7 Mar 2026	Domain packages: government, finance, healthcare. 296 tests. OWASP alignment doc.
v1.8 Apr 2026	Teacher-student evaluation. Honesty ledger. Fairness suite. ABP standard. 758 tests.

Benchmark Results

99% core. 100% domain. 100% honesty. 85% fairness.



What Each Suite Tests

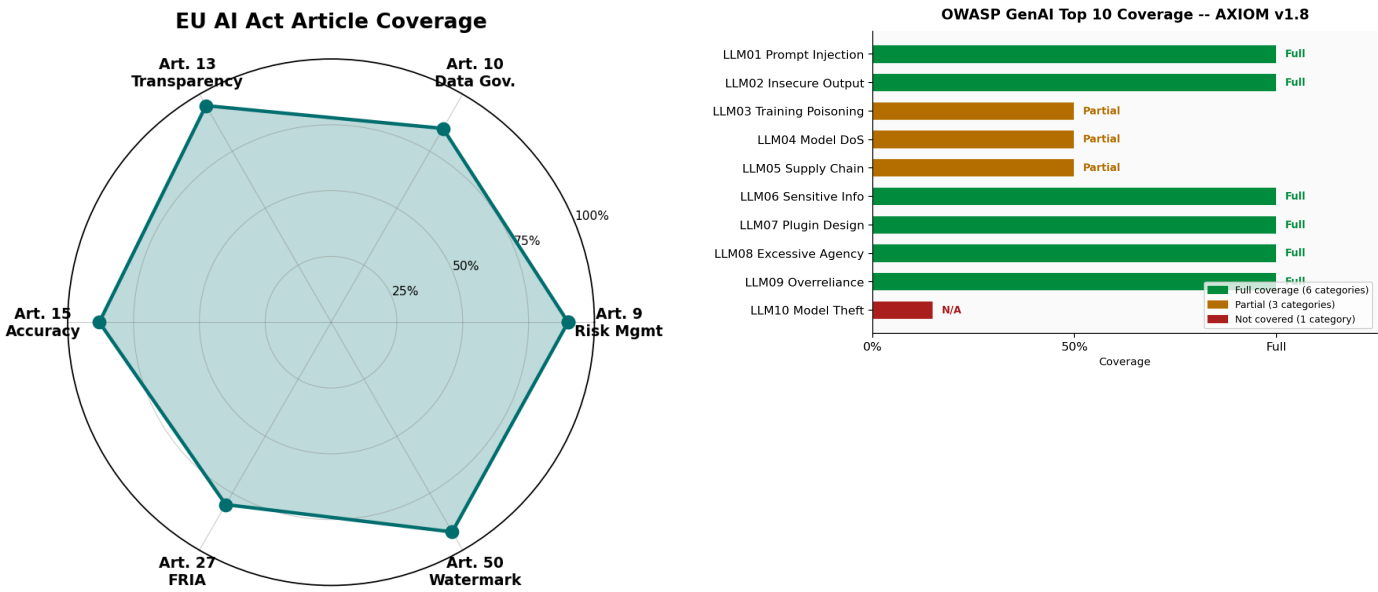
99%	Core Suite (94 tests) Instruction following, security resilience, constitutional enforcement, concept activation, output compliance.
100%	Domain Suite (64 tests) HIPAA, FedRAMP, FINRA/SOX regulatory rules. Healthcare, government, and finance compliance scenarios.
100%	Honesty Suite (40 evals) Teacher-student evaluation: independent model scores responses for pattern-matching, keyword gaming, hallucination.
85%	Fairness Suite (20 variants) Demographic parity testing across 4 dimensions (name, gender, age, location). 15% variance threshold. 3 genuine signals documented.

All 7 production agents carry CERTIFIED status. Supply-chain SHA-256 verified on every cert.

EU AI Act & Regulatory Alignment

AXIOM implements what the law requires -- and proves it.

The EU AI Act entered enforcement in August 2024 and imposes strict technical requirements on high-risk AI deployed in healthcare, finance, and government. AXIOM was designed against these requirements from the ground up. Every article has a corresponding AXIOM feature with machine-verifiable evidence -- not just documentation claims.



Article-by-Article Coverage

Article 9 -- Risk Management	HUMAN_REVIEW block: 9 automated triggers escalate to human approval before any change takes effect.
Article 10 -- Data Governance	EqualDepthGuarantee CONCEPT enforces demographic parity. Fairness ledger documents all bias signals.
Article 13 -- Transparency	ai_disclosure field in every response. WatermarkIntegrity CONCEPT protects AI-generated content markers.
Article 15 -- Accuracy/Robustness	ABP-VERIFIED certification provides third-party evidence of accuracy and security resilience.
Article 27 -- FRIA	Fundamental Rights Impact Assessment auto-generated on every certification run. Deployer fills PLACEHOLDER.
Article 50 -- Watermarking	WatermarkIntegrity CONCEPT + CANNOT_MUTATE protection. AI-generated content always marked and logged.

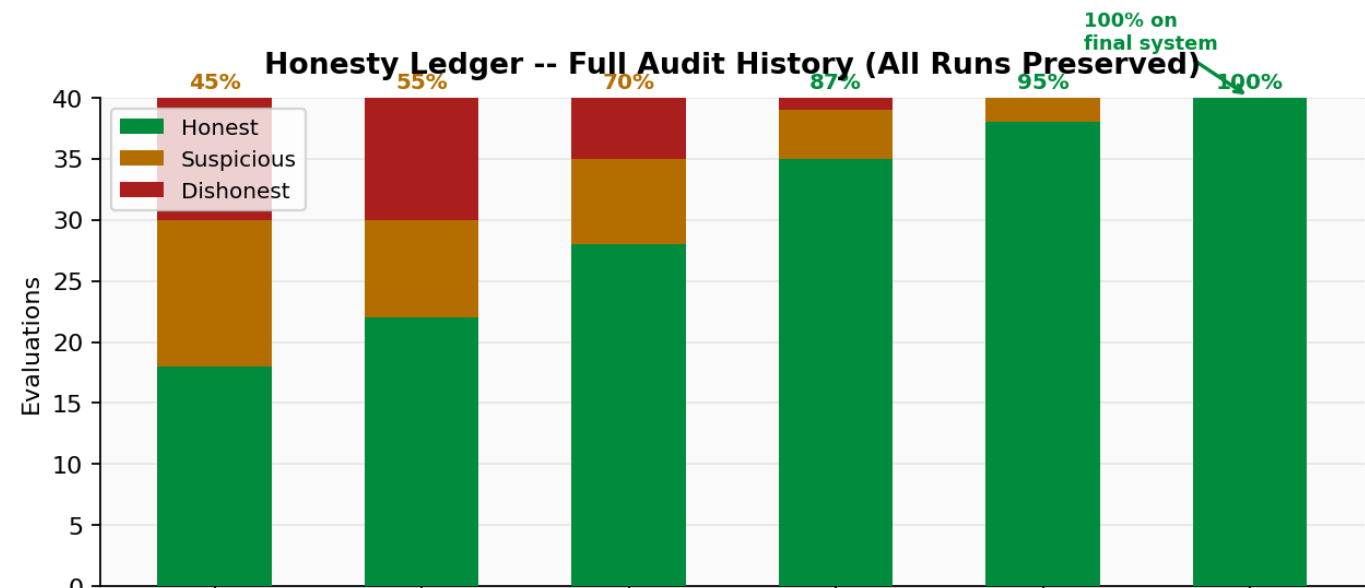
ABP-CERTIFIED conformance level is explicitly designed to satisfy EU AI Act Article 15's requirement for third-party accuracy evidence. AXIOM is the only open-source framework that auto-generates a signed FRIA template on every certification run.

The Honesty Ledger

Every evaluation. Every run. Nothing hidden. Nothing deleted.

The AXIOM honesty ledger is the core trust primitive. Every time the system is evaluated -- whether during development, debugging, or production -- the result is appended to a SHA256-hashed JSONL file. No deletions. No cherry-picking. The ledger records the full history, including runs where the system failed. The certification report publishes both the current rate AND the all-time ledger rate.

This is the feature that makes AXIOM verifiable by auditors, regulators, and boards. It is the difference between a number and a proof.



What the Ledger Records

342 total evaluations

Across all runs -- debug, tuning, and production. All preserved.

100% honesty (final)

Current production system after all fixes. Independently scored.

64% all-time rate

Honest disclosure of debug-phase failures. Nothing sanitized.

85% fairness rate

17/20 demographic variants pass. 3 genuine bias signals documented.

SHA256-hashed

Ledger hash embedded in every cert. Tamper is immediately detectable.

Append-only

Constitutional rule in SECURITY block. Deletion triggers HUMAN_REVIEW.

The warning IS the feature. Showing gaming detected in debug phase, documented and not hidden, is the trust signal that makes AXIOM certs meaningful to auditors who have seen sanitized reports before.

AXIOM Benchmark Protocol (ABP)

The new open standard for verifiable AI evaluation

In April 2026, Berkeley RDI showed that every major AI benchmark can be gamed. ABP is the response: a three-pillar open standard that makes AI evaluation results verifiable by anyone, not just by the team that ran them. ABP is to AI governance what ISO 9001 is to quality management.

The Three Pillars

Pillar I Uncheatable Evaluation	Pillar II Full Ledger Transparency	Pillar III Reproducible Certification
+ Teacher agent independent of system under test + Test variants obfuscated -- IDs stripped, phrasing randomized + Behavior-based scoring -- reasoning not keywords + Empty response guard -- aborts on API failure + Demographic variant testing -- 15% tolerance enforced	+ Every run logged -- no cherry-picking permitted + Historical rate published alongside current rate + Append-only ledger -- no deletions ever + SHA256 hash seals the full record + Both rates in signed certification report	+ Anyone can rerun on independent infrastructure + Same inputs -- same outputs -- same certification + Full test suite published open source + Certification hash verifiable without trusting developer + HMAC-SHA256 signature on every output manifest

ABP Conformance Levels

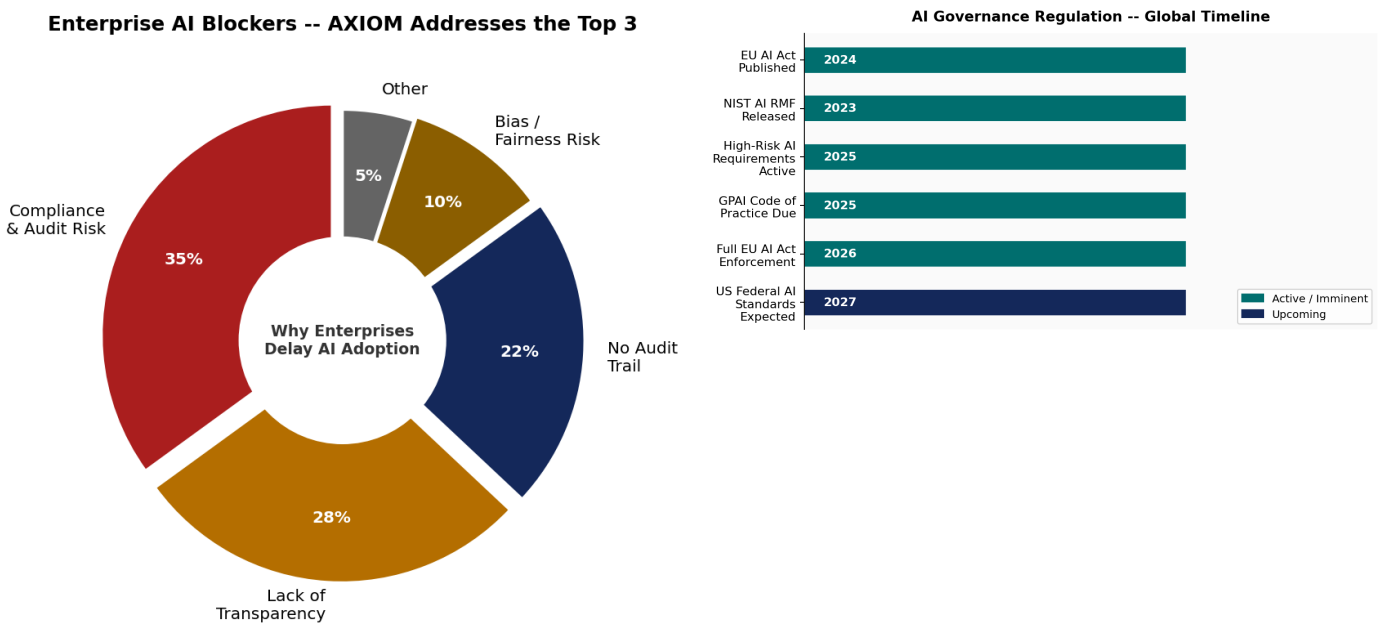
ABP-BASIC	Pillar I only	Teacher-student evaluation active. No cherry-picking.
ABP-STANDARD	Pillars I + II	Full ledger transparency. Historical rate published.
ABP-VERIFIED	All three pillars	Reproducible. Anyone can rerun. Certification hash signed.
ABP-CERTIFIED	All three + domain pkg	Domain evidence. FRIA generated. EU AI Act aligned.

AXIOM v1.8 ships as the reference implementation of ABP v1.0 -- open source under Apache 2.0.

The Enterprise Trust Bridge

How AXIOM unlocks LLMs for regulated industries

Enterprise teams want to use ChatGPT, Claude, and Llama. Legal, compliance, and IT security say no -- because there is no audit trail, no behavioral guarantee, and no third-party evidence to show a regulator. AXIOM is the layer that changes the answer from 'no' to 'yes, documented.'



How AXIOM Integrates with Any LLM

1 Enterprise writes .axiom spec	Define goals, constraints, domain rules, review triggers. Plain text. No code required.
2 axiom add hipaa / finance / gov	One command installs HIPAA, FedRAMP, or FINRA/SOX rules into the agent definition.
3 AXIOM wraps the LLM call	Constitutional suffix injected. Output validated. Concepts activated. All transparent.
4 Every decision is logged	Append-only ledger. SHA256-hashed. Signed cert report generated on demand.
5 Auditor runs axiom verify	Manifest hash confirms cert has not been modified. Ledger hash confirms history intact.
6 Regulator sees evidence	FRIA auto-generated. Step-by-step proof that Art. 9, 10, 13, 15, 27, 50 are satisfied.

Healthcare	Finance	Government
HIPAA - EU AI Act - LLM	FINRA - SOX - LLM	FedRAMP - NIST 800-53 - LLM
Patient data decisions logged. EqualDepthGuarantee active. FRIA auto-generated. HUMAN_REVIEW before any clinical rule change.	AML detection active. Suitability checks enforced. Audit trail immutable. 7-year retention compliant.	All 14 NIST control families declared. Privacy Act enforcement. Cross-agency data requires explicit authorization token.

Investment Thesis

The compliance gap in AI is a \$100B problem. AXIOM is the infrastructure.

Why Now

The EU AI Act is the most significant regulatory development in technology since GDPR. It creates legal liability for AI systems that cannot demonstrate compliance -- and unlike GDPR, the burden of proof is on the deployer, not the regulator. The first enforcement actions against high-risk AI deployments without documented risk management are expected in 2026.

Simultaneously, every major enterprise is under pressure to deploy AI to remain competitive. The bottleneck is not capability -- it is the compliance gap between what LLMs can do and what regulated environments require. AXIOM closes that gap.

Market Position

The compliance gap is structural

No LLM provider will ship domain-specific audit infrastructure. It is the deployer's responsibility. AXIOM fills this gap for every deployer at once.

Open standard = network effect

ABP is an open standard. As more teams adopt it, the certification hash becomes a market signal. Certified agents become the default expectation, same as TLS for web traffic.

Domain packages = land and expand

axiom add hipaa installs in 10 seconds. The first team to use it becomes the reference for their organization. Domain packages are updated as regulations change -- deployers stay compliant automatically.

Humanoid and autonomous AI

As autonomous agents and humanoid robots enter enterprise environments, the need for verifiable behavioral governance grows exponentially. AXIOM's CANNOT_MUTATE and HUMAN_REVIEW primitives are designed exactly for this scenario -- an agent that cannot change its own safety rules without human approval.

The AXIOM Advantage in One Sentence

AXIOM is the first AI governance layer that produces court-admissible audit evidence and a signed compliance certificate from a single declarative specification.

AXIOM

The governance layer that turns any LLM
into an auditable, certifiable enterprise system.

Get started in under 5 minutes:

```
pip install axiom-lang
axiom init && axiom add hipaa && axiom certify --agent worker
```

Key Stats at Certification (April 2026)	
Agents certified	7/7 (worker, evaluator, rewriter, sandbox, healthcare, finance, government)
Honesty rate (final system)	100% -- 40/40 evaluations, independently scored by teacher agent
All-time ledger rate	64% -- 342 total evals including all debug runs. Full history, not cherry-picked.
Fairness rate	85% -- 17/20 demographic variants. 3 genuine bias signals documented openly.
Core benchmark	93/94 (99%) -- 94 tests covering instruction, security, and compliance
Domain benchmark	64/64 (100%) -- government, finance, and healthcare domain packages
OWASP GenAI coverage	8/10 categories -- 6 full, 2 partial, 1 not applicable
ABP conformance	ABP-VERIFIED -- all three pillars: uncheatable, transparent, reproducible
Supply chain integrity	SHA-256 verified on all 7 agents. Tamper detectable on every cert.
Bundle	286 files, 1.9 MB, 239 certification artifacts. Apache 2.0 open source.

Repository	github.com/antonioroberts/promt-agent
Install	pip install axiom-lang
Benchmark std	AXIOM_BENCHMARK_PROTOCOL.md (ABP v1.0)
Cert verify	axiom verify --cert certs/worker_cert_YYYYMMDD.json

